SIM WARNING

Specification & Architecture

Requirements, acceptance scenarios and UML

Application version 1.0.0
Specification version 0.5 (15 October 2026)
Architecture diagrams reflect the current implementation

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Sim Warning — requirements and design specification

Version: 0.5, updated on 15 October 2026 with the approved manual-SIM fallback and first integrated test APK implementation.

This document records agreed requirements. Explicitly marked open questions are not approved behavior. The native setup, main/Settings/Help UI and Android monitoring integration now exist. This does not establish full device compliance: see TESTING.md and TDD_EVIDENCE.md for executed scope, sleep-delivery limitations, and remaining physical-device gates.

1. Purpose and scope

ID	Requirement
R01	Warn about unexpected loss of normal mobile network service, regardless of cause. Do not diagnose a loose SIM or imply a known cause.
R02	Support Android 8.0/API 26 and newer, using native Kotlin and Jetpack Compose as specified in the supplied blueprint.
R03	Keep this version minimal and entirely local: no account, advertising, analytics, or server connection.
R04	Monitor in the background without requiring an open screen. After initial setup, automatically resume on phone startup if monitoring was enabled. Preserve an explicit off setting across restarts.
R05	Minimize battery, memory, and CPU use. Prefer event-driven monitoring; avoid continuous polling, busy loops, and keeping the CPU awake throughout normal monitoring.

Normal service means mobile network service, not internet reachability. Emergency-calls-only service does not qualify. Wi-Fi does not qualify. Physical SIMs and eSIMs count equally.

2. Connectivity states and suppression

ID	Requirement
R06	Evaluate all SIMs. Any SIM with normal mobile service prevents a confirmed total outage; a loss on one SIM alone must not warn if another has service.
R07	Turning mobile data off alone must not warn while normal mobile service remains available. Wi-Fi must not conceal a mobile-service outage.
R08	Airplane mode and reliably identified intentional disabling of all SIMs suppress both warning types: clear their visual warnings and cancel pending sound/vibration. Do not infer intention merely from missing service or a disappearing SIM.
R09	Distinguish reliably disconnected service from unreadable connection status. Never report a confirmed outage solely because status cannot be read.
R10	Missing required permissions are a separate prerequisite state. Show “Monitoring needs permission” and “Review permissions”; do not claim active monitoring or issue repeated status-unavailable warnings. If a required permission is revoked, clear existing warnings and cancel all pending sound/vibration, including held alerts. Once permissions are restored, resume only if monitoring is enabled, starting fresh with the 5-minute initial grace period rather than resuming previous timers.

Approved first-test-APK fallback, 15 October 2026: public Android APIs cannot reliably identify manual all-SIM disabling on every phone. The user approved using the Monitoring switch before manually disabling all SIMs, with this explained in setup and Help. Airplane-mode suppression remains automatic. Never interpret an empty subscription list as proof of intention. This resolves the proposed fallback in ANDROID_FEASIBILITY.md for this test APK; automatic manual-disable recognition is not claimed.

3. Confirmed connection-loss timing

ID	Requirement
R11	During established monitoring, require 30 seconds of sustained loss before posting the visual warning. Shorter losses do not warn.
R12	Audible/vibration interval choices are 2, 4, 6, 8, or 10 minutes; default 2 minutes. For ordinary outages, measure deadlines from loss onset: I, 2I, 3I, etc.
R13	Use a fresh 5-minute initial grace period after phone restart, initial monitoring start, manual re-enabling, airplane-mode exit, or re-enabling SIM service after intentionally disabling all SIMs, if service is unavailable. Do not resume old outage timers.
R14	For these initial episodes, post the visual warning and issue the first sound/vibration at 5 minutes if service is still unavailable. Repeat at the chosen interval from that deadline: 5 minutes, 5+I, 5+2I, etc. No outage warnings occur during the initial grace period.
R15	Interval changes apply immediately without resetting the episode's timing anchor. For ordinary outages, retain loss onset. For initial episodes, preserve the first-warning deadline at 5 minutes and measure subsequent repeats from that deadline. Use the next future deadline under the new interval, with no catch-up sounds.
R16	Confirm recovery only after 30 seconds of continuous normal mobile connectivity. Then clear warnings and cancel pending alerts quietly, without a recovery sound or separate recovery notification.
R17	A shorter reconnection does not reset the outage timer or clear the active warning. Hold any sound/vibration due during recovery confirmation. On confirmed recovery cancel it; if recovery fails, issue the held warning once, subject to current system settings, then continue the original schedule.

ID	Requirement
R18	Switching monitoring off clears both warning types and cancels all pending sounds/vibration, including held warnings. Re-enabling starts fresh under R13.

During any initial episode under R13, 30 continuous seconds of normal mobile connectivity completes startup/reconnection confirmation and ends the initial grace period early. Cancel initial-episode warning timers. Any subsequent loss is an ordinary outage under R11–12.

For an ordinary loss still in its initial 30-second grace period, a brief reconnection preserves the original loss onset. Hold new warning presentation while the 30-second recovery check is in progress. If recovery succeeds, cancel the episode without warning. If recovery fails after the original loss grace deadline, post the visual warning immediately; preserve the original sound/vibration schedule and apply R17 to any held alert. Existing visual warnings remain posted during recovery confirmation.

Timing examples, assuming uninterrupted loss and sound/vibration permitted:

Episode	Selected interval	Visual warning	Sound/vibration deadlines
Ordinary loss	2 minutes	30 seconds	2, 4, 6, 8 minutes…
Ordinary loss	6 minutes	30 seconds	6, 12, 18 minutes…
Startup or exit from intentional disconnection	2 minutes	5 minutes	5, 7, 9, 11 minutes…
Startup or exit from intentional disconnection	6 minutes	5 minutes	5, 11, 17 minutes…

Exact boundary ordering and delivery tolerance require O08 and technical validation; these times are intended deadlines, not verified delivery guarantees.

4. Connection status unavailable

ID	Requirement
R19	During established monitoring, after 30 seconds of continuously unreadable status, post a separate status-unavailable notification without requiring the app to be open. For initial episodes covered by R13, use a 5-minute visual grace period instead.
R20	First issue its distinct sound/vibration after 10 continuous minutes, then every 10 minutes: 10, 20, 30 minutes, etc. This interval is independent of the user's outage interval.
R21	If a confirmed outage becomes unreadable, replace the outage state with status unavailable, cancel its sound/vibration schedule and held warnings, and start the unreadable timer. Never run both schedules concurrently.
R22	Once status is reliably readable, cancel the unreadable-state alert schedule. If connected, quietly confirm recovery using the 30-second rule. If disconnected, start the ordinary loss rules, subject to intentional suppression and any applicable initial grace period.
R23	Intentional-disconnection suppression, monitoring off, and system sound/vibration rules also apply to status-unavailable warnings.

5. Notifications, sounds, and vibration

ID	Requirement
R24	Post visual warnings independently of sound suppression. Android permissions and user-controlled visual notification settings still govern actual display. Do not bypass those settings.
R25	Respect Do Not Disturb rules. Respect silent mode for sound and vibration; in vibrate mode allow vibration subject to system settings, but no sound. Never alter these settings to warn.
R26	When suppression ends, wait for the next scheduled warning; do not replay missed warnings. Monitoring and visual posting continue while sound is suppressed.
R27	Dismissing a confirmed-outage notification does not stop monitoring or repeating alerts. Repost the visual notification at the next scheduled warning. The repost itself produces no extra sound or vibration; the independent scheduled alert still follows R25.
R28	Tapping the warning opens the main screen with connection status and access to Settings. Show localized elapsed outage duration below the connection-loss title, using a battery-efficient mechanism.
R29	Use the two approved original sounds: connection loss uses the 2.1-second rounded three-note motif in design/connection-loss-candidate.wav (523.25, 440.00, 587.33 Hz); unavailable status uses the 1.8-second lower descending two-note motif in design/status-unavailable-candidate.wav (392.00, 329.63 Hz). Preserve their reviewed audio, timing, and distinct timbres. Their candidate filenames do not indicate pending approval.
R30	Vibrate at scheduled warning deadlines and apply the same hold/cancel timing rules as sound. Visual-only posting or reposting must not itself vibrate.

Approved vibration design: three short pulses for connection loss and two short pulses for unavailable status, matching each sound's note count. Exact pulse lengths, gaps, and supported-device feel remain to tune and review; system suppression rules still apply.

Status-unavailable notifications use the same tap behavior as connection-loss notifications. Dismissing one does not stop monitoring or its 10-minute warning schedule. Repost it silently at the next status-unavailable warning deadline, with no extra sound/vibration caused by reposting; separately scheduled sound/vibration still follows system settings.

Approved localized text:

Text	English	Danish	German
App name	Sim Warning	SIM-advarsel	SIM-Warnung
Loss warning	No mobile network connection	Ingen forbindelse til mobilnettet	Keine Verbindung zum Mobilfunknetz
Unreadable status	Connection status unavailable	Forbindelsesstatus er ikke tilgængelig	Verbindungsstatus nicht verfügbar

6. Setup, settings, localization, and accessibility

ID	Requirement
R31	First-time setup has four short steps: language, brief explanation, warning interval, required permissions. Enable monitoring on completion when prerequisites are granted.

ID	Requirement
R32	Offer Danish, English, and German during setup and in Settings. Show native language names with flags: Denmark/Dansk, UK/English, Germany/Deutsch. Preselect the supported phone language, otherwise English; allow a change before continuing.
R33	Use stable text IDs and per-language JSON catalogs such as TextCatalog_DEN.json, TextCatalog_ENG.json, and TextCatalog_GER.json for all app-owned text. Include setup, help, settings, statuses, permission explanations, notifications, and duration formatting. Android-owned dialogs retain system translations.
R34	Require only essential permissions, justified individually against functionality. Provide simple initial granting and later review/grant access through the app, invoking Android dialogs/settings as appropriate.
R35	Main screen: connection status, monitoring switch, Settings button. Show the contextual permission-review action when required.
R36	Settings has six items: Language, Warning interval, Preview sound, Test warning, Permissions, Help. Use familiar Android navigation and normal back behavior.
R37	Preview sound and Test warning each offer both warning types and respect current system settings. Test warning uses a clearly marked sample notification and allowed sound/vibration; it must not alter live state/timers or replace a live warning.
R38	Help uses friendly, clear, respectful language without being patronizing. Explain usage, grace periods, warning intervals, sound modes, intentional-disconnection suppression, and permissions. Apply that tone throughout all languages.
R39	Approved sizing direction: 18sp body text, approximately 28sp main headings, controls at least 48dp high, and Android text scaling. Validate all three languages at enlarged sizes without clipping.
R40	Modern Nordic style: white dominant, medium blue and black secondary, generous spacing and clear readability. Reviewed main-screen direction approved. Approved logo: “Nordic Signal”, a medium-blue (#2769AD) SIM-card outline with rounded rising signal bars on white. Preserve its reviewed proportions and shapes; launcher/adaptive and monochrome notification variants require technical preparation and small-size verification.

Approved logo assets: editable SVG and PNG preview. The same symbol is used across all languages; localized app names remain unchanged.

ID	Requirement
R41	Implement the approved four-step setup design described in UI_SETUP_SPEC.md, including its layout, localized content, navigation, interval selection, and permission-gated completion. The preview's simulated permission handling is not production behavior.
R42	Give app-owned screens and interactive/status elements stable, language-independent automation IDs. Use Compose test tags, expose them to UI Automator through resource-ID semantics, and preserve accessible human-readable labels separately. Follow the element registry and test contract in UI_SETUP_SPEC.md.
R43	Make setup UI tests runnable on an Android emulator and on a physical device. Use controlled connectivity/permission inputs for deterministic app UI tests and real Android dialogs in separate integration tests; never ship test-only permission or connectivity overrides in release builds.
R44	Implement the approved Settings menu and detail-screen design in UI_SETTINGS_SPEC.md, including stable element IDs and emulator/device acceptance scenarios. Keep preview simulations separate from actual audio, notifications, permission state, and live monitoring.
R45	While the monitoring service is running, provide a quiet “Monitoring is on” service notification, localized through the text catalogs and separate from both warning types. It must not itself play sound or vibrate. Tie its presence and wording to actual service/readiness state rather than the saved enabled preference alone; system notification settings still govern presentation. Remove it when monitoring stops.

7. Acceptance scenarios

These scenarios define intended behavior, not completed tests. Each must be translated into tests before its production implementation, per the supplied TDD directive.

ID	Given / when	Expected outcome	Requirements
A01	Ordinary loss lasts 29 seconds	No warning	R11
A02	Ordinary loss persists beyond 30 seconds; interval 2 minutes	Visual at grace expiry; sound/vibration at 2, 4, 6 minutes subject to settings	R11–12
A03	Startup has no service; interval 2 minutes	No outage warning before 5 minutes; visual and first alert at 5, repeats at 7, 9 minutes	R13–14
A04	Leave airplane mode or re-enable SIM service with no service	Fresh 5-minute initial timing, no old timers resumed	R08, R13–14
A05	One SIM loses service while another remains connected	No total-outage warning	R06
A06	All SIMs have only emergency service	Treat as no normal service	R01, R06
A07	Mobile data off but normal service remains	No outage	R07
A08	Wi-Fi connected but all mobile service lost	Apply normal outage warning rules	R07
A09	Airplane mode or confirmed intentional all-SIM disabling begins	Clear either warning type; cancel all associated pending alerts	R08, R23
A10	Recovery lasts 29 seconds then fails	Preserve outage timer and warning; release held alert once if one became due	R16–17
A11	Recovery remains continuously available for 30 seconds	Quietly clear warning and cancel alerts	R16–17
A12	Sound/vibration deadline falls during recovery confirmation	Hold; cancel if confirmed, otherwise release once under current settings	R17, R30
A13	DND suppresses warnings	Visual still posted; suppressed outputs do not play	R24–26
A14	Phone silent / vibrate mode	Silent: no sound/vibration. Vibrate: no sound, permitted vibration only	R25
A15	DND or silent mode ends between deadlines	No catch-up; use next scheduled deadline	R26
A16	Ordinary outage interval changes from 4 to 2 minutes at minute 5	No immediate catch-up; next alert at minute 6	R15
A17	User dismisses outage notification	Keep monitoring; silent visual repost at next alert deadline	R27
A18	Tap warning	Open main screen; duration localized	R28
A19	Monitoring switched off during either warning state	Clear warning and cancel all sound/vibration; retain off after reboot	R04, R18
A20	Re-enable monitoring without service	Fresh 5-minute initial grace	R13, R18

ID	Given / when	Expected outcome	Requirements
A21	Status continuously unreadable	Visual after 30 seconds; distinct sound/vibration at 10, 20 minutes	R19–20
A22	Confirmed outage changes to unreadable	Stop outage schedule; start unreadable schedule; no concurrent schedules	R21
A23	Unreadable status becomes readable	Cancel unreadable schedule and follow detected state	R22
A24	Required permission missing or denied	Permission-needed status/action; no repeated unreadable-status alerts or false active claim	R10, R34
A25	First launch in each supported phone language; unsupported language	Appropriate preselection, English fallback; choice remains editable	R31–33
A26	Review all screens with large text in each language	Readable, usable controls, no clipping or untranslated text	R33, R38–40
A27	Preview/test either warning type during a real outage	Correct sample output subject to settings; live state and notification unaffected	R29, R37
A28	Initial episode interval changes from 2 to 4 minutes before minute 5	First warning remains at minute 5; subsequent alerts at minutes 9, 13, etc.	R14–15
A29	Initial episode interval changes from 4 to 2 minutes at minute 8	No immediate catch-up; next alert at minute 9, retaining the minute-5 anchor	R14–15
A30	Status remains unreadable from startup or another initial episode under R13	Visual notification after 5 minutes; first sound/vibration at 10 minutes, then 20, 30 minutes, etc.	R13, R19–20
A31	Normal service becomes available during an initial episode and remains available for 30 seconds	End initial mode and cancel its timers; a later loss uses the ordinary 30-second visual grace and onset-based alert schedule	R11–14, R16
A32	Ordinary loss at second 0; service returns at second 20 and fails at second 40	No warning during the recovery check; post visual warning at second 40. Keep loss onset at second 0 for alert timing	R11–12, R16–17
A33	Ordinary loss at second 0; service returns at second 20 and remains available through second 50	No warning while recovery is checked; cancel episode at second 50	R11, R16–17
A34	Required permission revoked during either warning state	Clear active warnings and cancel pending/held sound and vibration; show permission-needed status	R10
A35	Required permissions restored while monitoring is enabled and service remains unavailable	Fresh 5-minute initial grace; no old warning timers resumed	R10, R13–14
A36	Required permissions restored while monitoring is off	Remain off; do not start monitoring or warning timers	R04, R10, R18
A37	Tap a status-unavailable notification	Open the main screen with current status and access to Settings	R28
A38	Dismiss a status-unavailable notification during ongoing unreadable status	Keep monitoring and the 10-minute schedule; silently repost at the next deadline with no extra sound/vibration from reposting	R20, R23–27

ID	Given / when	Expected outcome	Requirements
A39	Run the setup automation scenarios U01–U14 in UI_SETUP_SPEC.md	Stable selectors, correct navigation/state/localization, real permission gating, and accessible layouts across configured emulator/device cases	R31–43
A40	Run Settings scenarios S01–S12 in UI_SETTINGS_SPEC.md	Approved navigation, persisted choices, stable IDs, isolated previews/tests, localized accessible layouts, and real prerequisite handling	R36–44
A41	Monitoring service starts and later stops	Separate localized service notification while running, no sound/vibration from service-status posting; remove on stop without affecting the separate warning contracts	R04, R18, R45

8. Engineering validation and definition of done

Research and proposed architecture: ANDROID_FEASIBILITY.md, checked against official Android/AOSP documentation on 15 October 2026. Important unresolved constraints include idle-mode timing, foreground-service notification/type, and reliable recognition of intentional SIM disabling. This research does not change agreed behavior or establish device-tested reliability.

The supplied blueprint requires interfaces and tests before functional implementation. Pure Kotlin state/timing tests should exercise agreed scenarios with controlled time. Android component/UI tests must verify permission flows, text/layout, and notifications. Physical-device end-to-end tests must verify startup, screen-off operation, SIM transitions, suppression, and recovery. A feature is not fully verified until the applicable tiers pass; missing device access or unrun checks must be reported explicitly.

Respect the supplied low-memory development constraint: constrain build workers/daemons and verify available memory before choosing build settings. UI automation must support an emulator as requested; use a physical device when host memory cannot support it. Emulator tests do not replace real-device telephony and battery validation. No tools are installed as part of this specification update.

Before reliability claims, validate:

- Minimal permissions, Android-version-compatible telephony APIs, and subscription aggregation.
- Reliable recognition of intentionally disabled SIMs versus disappearance/read failure.
- Boot/first-unlock behavior, force-stop limitations, permission revocation, and device battery restrictions.
- A legitimate background-service design, any ongoing notification, and screen-off alert scheduling.
- DND visual suppression, channel sound/vibration settings, notification dismissal, and locale changes.
- Measurable power/memory/CPU use and acceptable timer delivery tolerance. Targets remain to agree.
- Matching catalog IDs, placeholder formatting, localization fallback, and packaged app-name behavior.

9. Remaining review points

ID	Open point
O07	Remaining palette details, exact vibration pulse/gap timing, and remaining localized copy. Settings, setup, both warning sounds, vibration pulse counts, and logo are approved; Android asset packaging, exact permission list, and system dialogs await engineering preparation/validation.

ID	Open point
O08	Exact simultaneous-event ordering, warning delivery tolerance, measurable resource targets, supported test-device matrix, and technical feasibility of Android background behavior.
O09	Persistent monitoring notification, process restart handling, and what users should see if Android prevents monitoring.

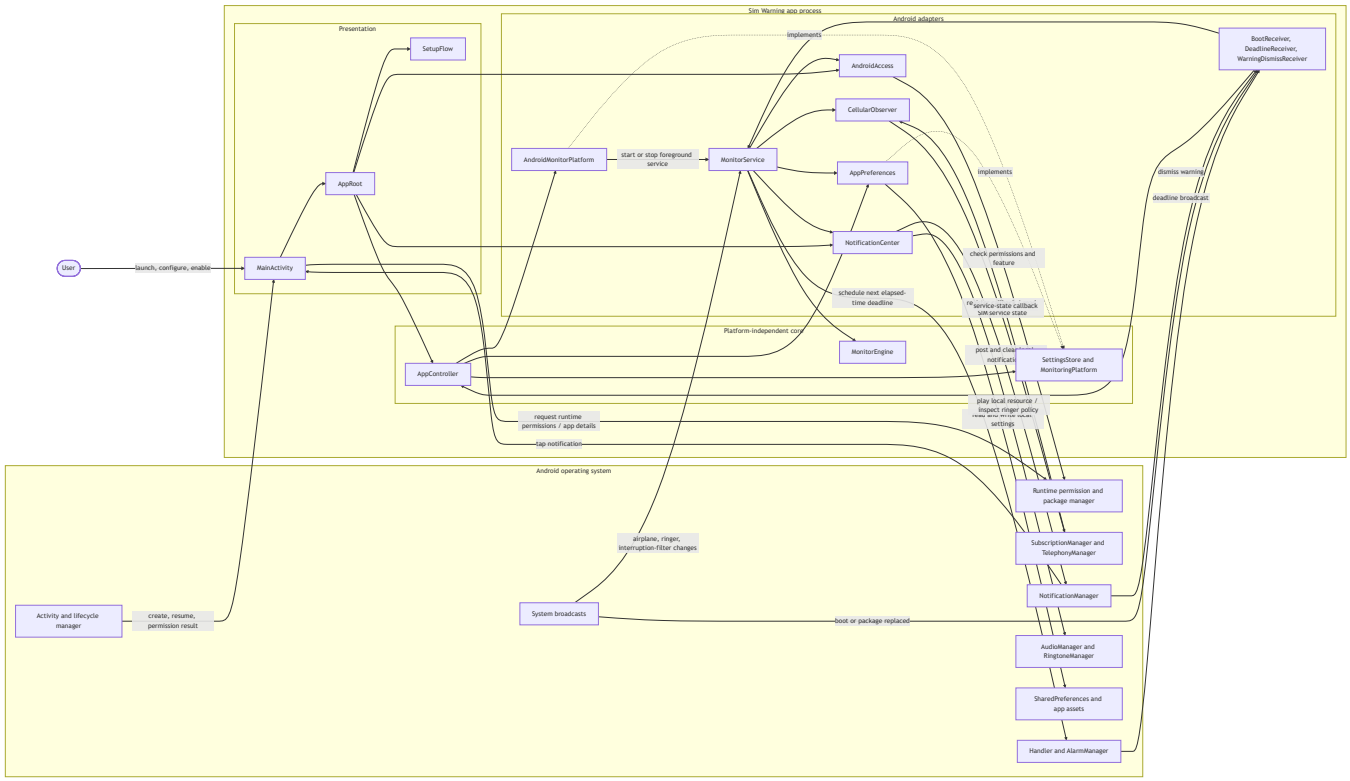
Resolve these interactively, one point at a time. Do not silently turn proposed behavior or unverified platform assumptions into requirements.

Application Architecture UML

These diagrams describe the current Android implementation, including the boundary between app-owned logic and Android system services. The diagrams are intentionally split by concern: component boundary, class structure, user setup, live monitoring, boot recovery, and monitor states.

Component And System Boundary

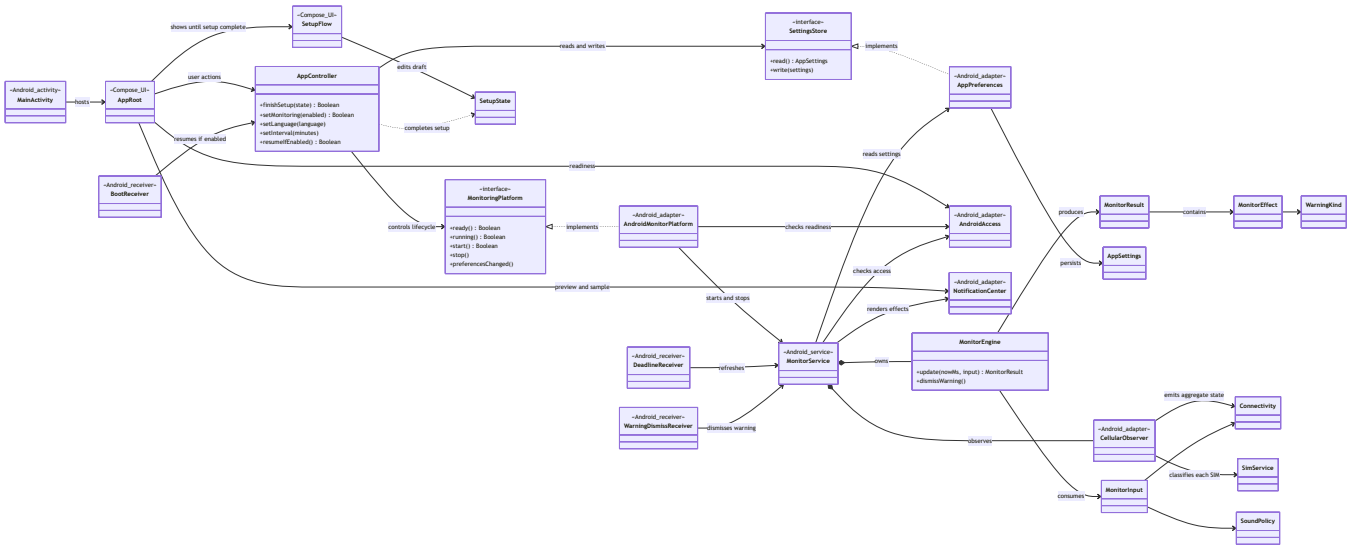
The app has no backend or internet endpoint. Android provides telephony observations, lifecycle/broadcast delivery, permissions, local persistence, alarms, and notification/audio facilities. Cellular registration is observed through Android's telephony API; the app does not open a network connection.



`MainActivity` is the launcher entry point. `MonitorService` and the receivers are app-private Android components; only the launcher activity is exported. System callback directions above represent Android API calls and callbacks, not app-to-app or internet communication.

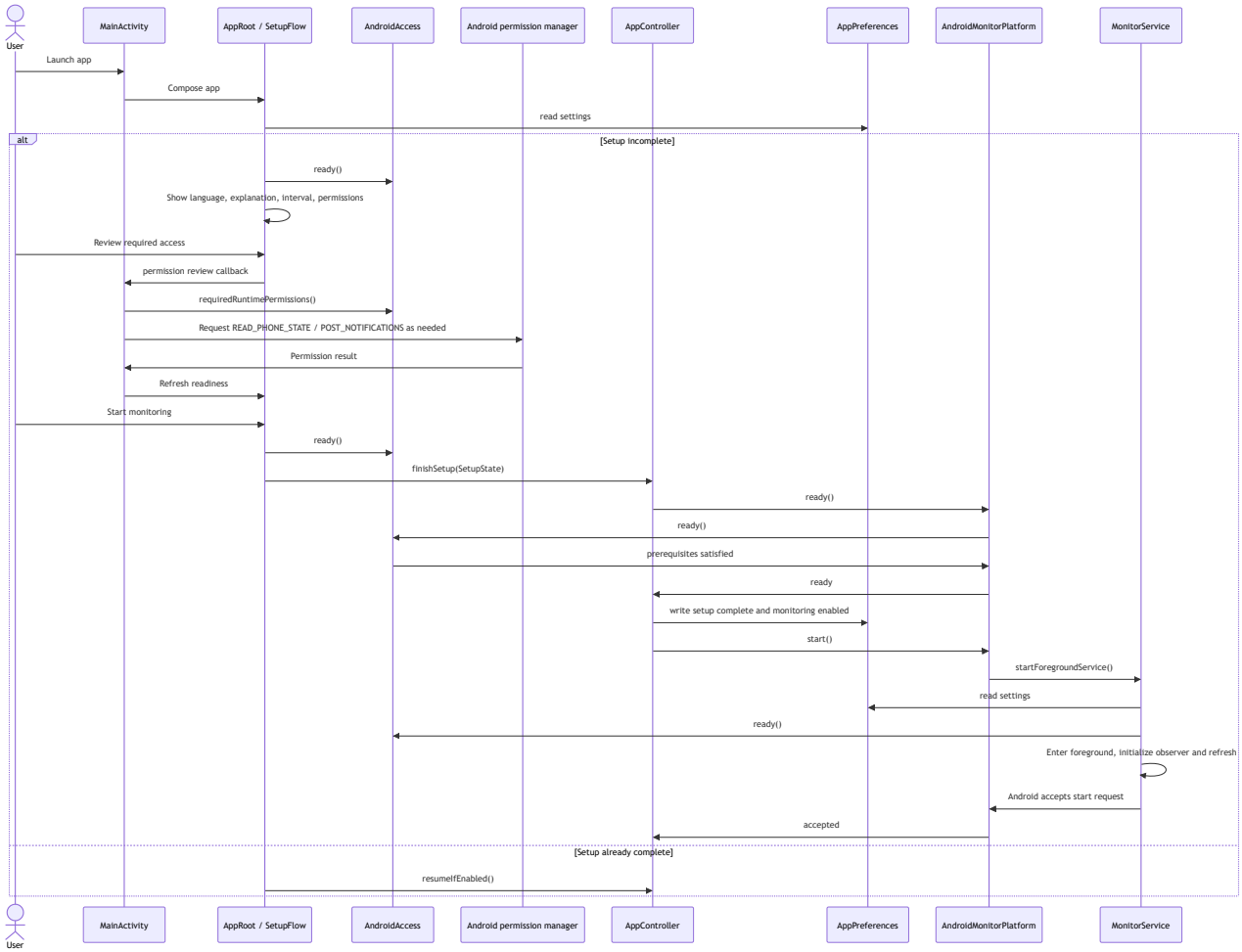
Class Structure

The core owns settings coordination and outage decisions. Android implementations satisfy the core interfaces and translate between Android APIs and core data types.



Setup And Monitoring Start

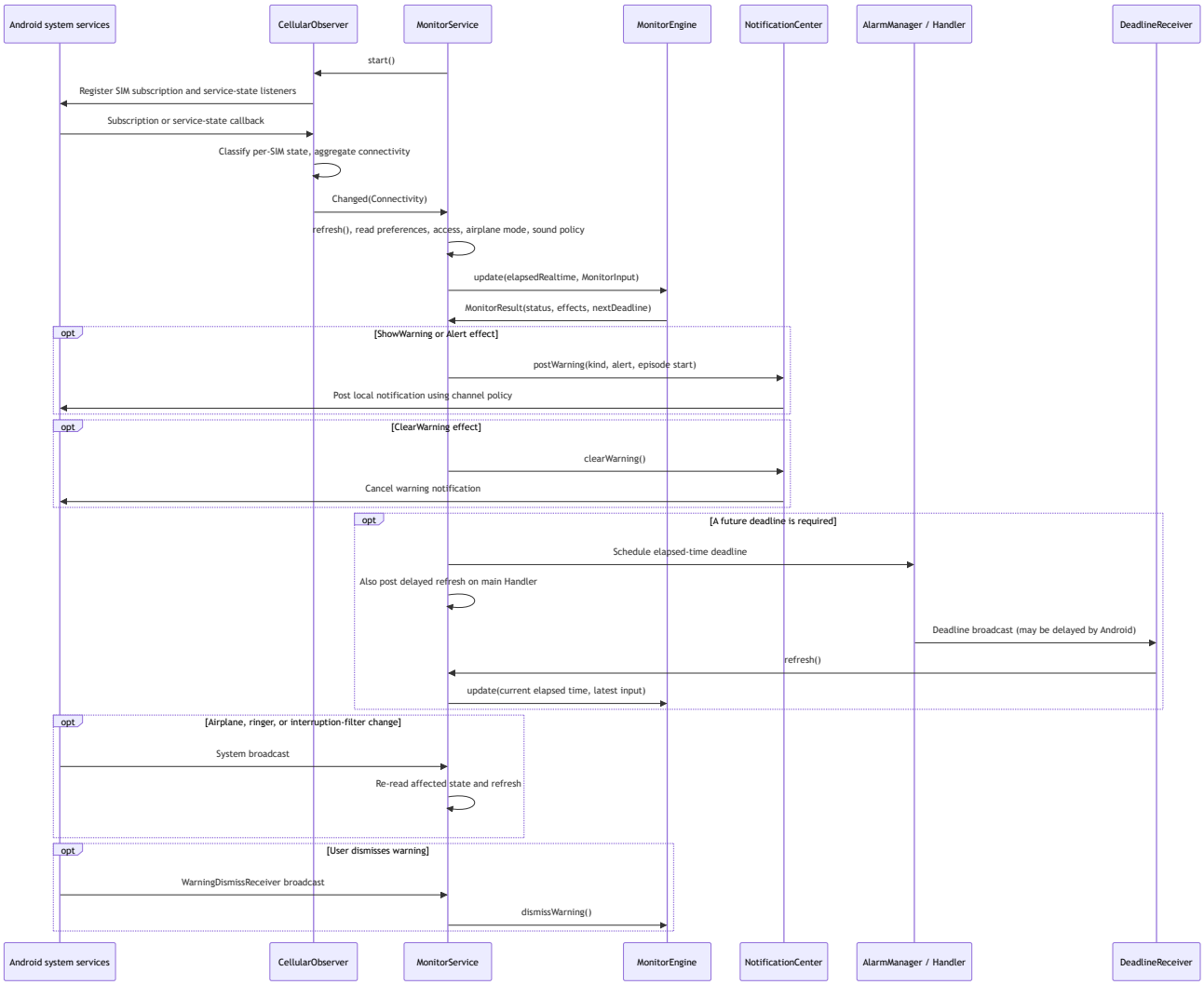
Permissions are requested by the activity through Android's runtime permission flow. `AndroidAccess.ready()` is rechecked before setup completion and before service start. `AppController` persists settings transactionally around the start request and restores the prior settings if startup fails.



The start result means Android accepted the start request, not that monitoring has remained active indefinitely. The foreground service independently validates state at creation and on subsequent refreshes.

Live Monitoring And Warning Delivery

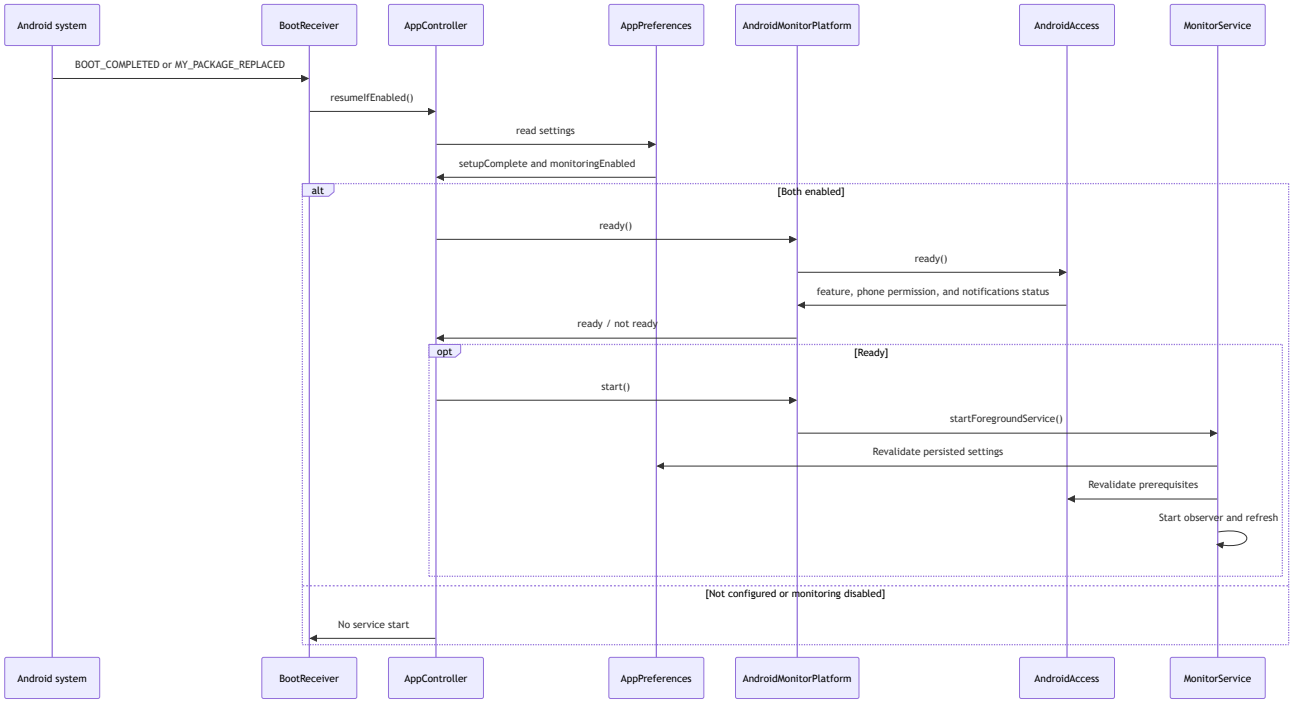
Monitoring is event/deadline-driven. The service submits each observation and current settings to the core engine; the engine returns a status, effects, and at most the next required deadline. The adapter applies effects locally and schedules that deadline without periodic polling.



Telephony observations concern cellular registration, not Wi-Fi or general internet connectivity. Audio uses bundled sound resources; the app does not request exact-alarm or DND-policy access.

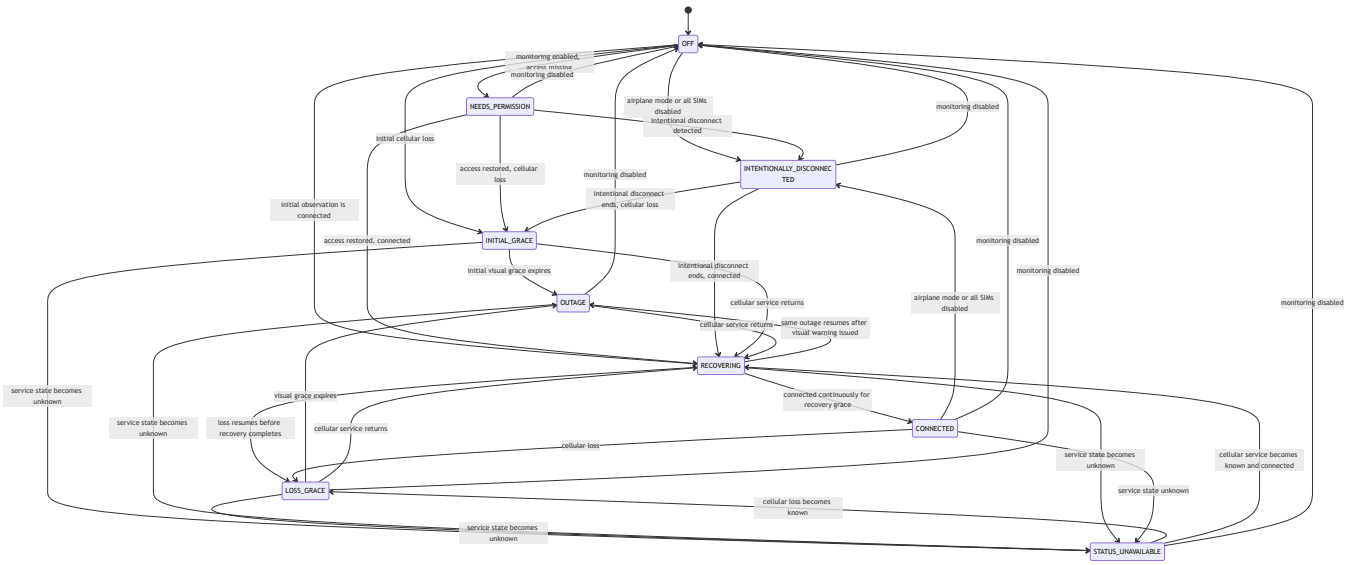
Boot And Package-Update Recovery

When Android delivers `BOOT_COMPLETED` or `MY_PACKAGE_REPLACED`, the private boot receiver asks the same controller/platform path to resume only if setup and monitoring are enabled and prerequisites remain available.



Monitor State Machine

This view summarizes externally visible `MonitorStatus` values. It omits internal episode timestamps and sound/vibration policy details; `MonitorEngine` remains the authoritative transition logic.



`INITIAL_GRACE` uses the initial five-minute visual grace. Later known connection-loss episodes use the configured warning interval as their grace. Recovery must remain connected for 30 seconds before the episode is cleared. Unknown service state is represented separately and uses its own ten-minute alert cadence.

Source Map

- Core contracts and effects: `MonitorContract.kt`
- Setup and lifecycle coordination: `AppController.kt`

- Outage state machine: `MonitorEngine.kt`
- Android component declarations and permissions: `AndroidManifest.xml`
- UI and permission entry points: `MainActivity.kt`, `AppRoot.kt`, `SetupFlow.kt`
- Service lifecycle and receivers: `MonitorService.kt`
- Telephony and Android adapters: `CellularObserver.kt`, `AndroidAccess.kt`, `AndroidMonitorPlatform.kt`
- Local persistence and notifications: `AppPreferences.kt`, `NotificationCenter.kt`